

Infrastructure Cost Forecast — Year 1

Created: 2026-02-28 **Updated:** 2026-02-28 — Switched production Azure Functions from EP1 (Elastic Premium) to Y1 (Consumption) to reduce costs. Annual saving of ~£1,488. **Currency:** GBP (£). Azure bills in USD; estimates use £1 = \$1.27 (approximate). All figures are estimates and should be re-validated against current Azure pricing at time of provisioning.

Assumptions

- **Launch timeline:** Q1 = soft launch (single retailer, curated creator cohort ~50); Q2 = growth (AWIN live, multiple retailers); Q3 = payouts live, scaling; Q4 = established product
- **Region:** Azure UK South (all resources), consistent with current Terraform config
- **Team size:** 2-3 developers
- **Traffic profile:** low but real from day one; meaningful growth from Q2; no viral spike scenario planned for (handle separately if needed)
- **Included features by quarter:** Q1 core MVP; Q2 AWIN affiliate integration + Redis caching; Q3 Stripe creator payouts + AWIN reconciliation; Q4 sustained growth
- **Not included:** VAT (add 20% for UK business costs where applicable); exchange rate fluctuation risk

Environment Architecture

Two permanent environments, matching current Terraform structure:

Environment	Purpose	Key difference
Dev/Staging	CI/CD deployments, QA, integration testing	Scale-to-zero, smaller SKUs, email dry-run
Production	Live users	Larger SKUs, GRS storage, 90-day log retention; Functions use Consumption plan (cold starts acceptable for async workloads)

The Container Registry (Basic SKU) is shared between both environments.

Part 1: Azure Infrastructure — Dev/Staging

Dev/Staging costs are relatively stable throughout the year. Redis is added in Q2 once the caching implementation is complete.

Service	SKU / Config	Monthly (Q1)	Monthly (Q2+)	Notes
Container App (API)	0.5 vCPU, 1Gi, scale-to-zero	£7	£7	Charged only when running; CI deployments drive most usage

Service	SKU / Config	Monthly (Q1)	Monthly (Q2+)	Notes
PostgreSQL Flexible Server	B_Standard_B1ms, 32GB, LRS	£14	£14	Dev DB; burstable 1 vCore
Azure Functions	Y1 Consumption	£1	£1	Scale-to-zero; dev traffic is negligible
Storage Account	Standard LRS	£2	£2	Media blobs + processing queue
Log Analytics Workspace	30-day retention	£3	£3	Dev logs; typically stays within 5GB free tier
Key Vault	Standard, ~20 secrets	£1	£1	Negligible ops cost
Container Registry	Basic SKU (shared with prod)	£4	£4	One registry; charged once
Azure Cache for Redis	—	—	£13	Basic C0 (250MB); added Q2
ACS Email	Dry-run mode	£0	£0	No emails sent in dev
Dev/Staging Total		£32/month	£45/month	

Annual Dev/Staging: (£32 × 3) + (£45 × 9) = £96 + £405 = £501

Part 2: Azure Infrastructure — Production

Production scales across four quarters as the user base and feature set grows.

Service Sizing by Quarter

Service	Q1 (M1–3)	Q2 (M4–6)	Q3 (M7–9)	Q4 (M10–12)
Container App	1 vCPU, 2Gi, min 1	1 vCPU, 2Gi, min 1	1 vCPU, 2Gi, min 1–2	2 vCPU, 4Gi, min 1–2
PostgreSQL	D2s_v3, 64GB, GRS	D2s_v3, 64GB, GRS	D4s_v3, 128GB, GRS	D4s_v3, 128GB, GRS
Functions	Y1 Consumption	Y1 Consumption	Y1 Consumption	Y1 Consumption
Redis	—	Standard C1 (1GB)	Standard C1 (1GB)	Standard C2 (6GB)
Storage	~20GB GRS	~50GB GRS	~100GB GRS	~200GB GRS
CDN	—	Basic	Standard	Standard

Monthly Cost Breakdown by Quarter

Q1 — Soft Launch (months 1–3)

Service	Monthly	Notes
Container App (API)	£57	1 vCPU, 2Gi, min 1 replica always on; ~£75 including peak scaling
PostgreSQL D2s_v3 + 64GB GRS	£80	~\$92/hr compute + \$0.115/GB storage + GRS backup
Azure Functions Y1	£3	Consumption plan; negligible executions during soft launch; 10–30s cold start on first media job after inactivity
Storage GRS ~20GB	£3	Hot tier, geo-redundant
Log Analytics + App Insights	£10	90-day retention; within or just above 5GB free tier
Key Vault	£2	~20 secrets + low ops volume
ACS Email (~3,000/month)	£1	Within or just above 3,000/month free tier
Q1 Monthly Total	£156	

Q2 — Growth Phase (months 4–6) ← Redis and CDN added; AWIN live

Service	Monthly	Notes
Container App (API)	£80	Increased traffic from AWIN clicks
PostgreSQL D2s_v3 + 64GB GRS	£80	Same SKU; no upgrade needed yet
Azure Functions Y1	£5	AWIN reconciliation job added; still low execution volume on Consumption plan
Redis Standard C1 (1GB)	£63	Feed caching, session data; Standard tier for HA/SLA
Storage GRS ~50GB	£5	Growing media library
CDN	£8	Azure CDN Standard for static assets + media delivery
Log Analytics + App Insights	£12	Slightly more telemetry with AWIN
Key Vault	£2	Additional AWIN API token secret
ACS Email (~6,000/month)	£1	Verification, moderation, onboarding emails
Q2 Monthly Total	£256	

Q3 — Scaling Phase (months 7–9) ← Stripe payouts live; DB upgraded; creator base growing

Service	Monthly	Notes
Container App (API)	£100	Average 1.5 replicas as shopper traffic grows

Service	Monthly	Notes
PostgreSQL D4s_v3 + 128GB GRS	£157	⚠ Upgrade trigger: read latency or CPU > 70% sustained
Azure Functions Y1	£8	3 active functions (media processing, AWIN reconciliation, payout confirmation); growing execution volume
Redis Standard C1 (1GB)	£63	Same tier; monitor memory usage
Storage GRS ~100GB	£8	Media growing with creator cohort expansion
CDN	£15	Growing asset delivery volume
Log Analytics + App Insights	£15	More services generating telemetry
Key Vault	£3	Stripe secrets + AES encryption key added
ACS Email (~10,000/month)	£2	Payout notifications added to email volume
Q3 Monthly Total	£371	

Q4 — Established (months 10–12) ← *Full feature set; sustained growth*

Service	Monthly	Notes
Container App (API)	£130	Average 1.8 replicas; may need 2 vCPU if CPU-bound
PostgreSQL D4s_v3 + 128GB GRS	£157	Same SKU; add read replica if query load warrants it (+ £145/month)
Azure Functions Y1	£12	Consumption costs grow with execution volume; still well under EP1 cost
Redis Standard C2 (6GB)	£133	⚠ Upgrade trigger: C1 memory utilisation > 70%
Storage GRS ~200GB	£13	~200GB accumulated media
CDN	£20	Standard volume
Log Analytics + App Insights	£20	90-day retention of growing telemetry
Key Vault	£4	Low growth
ACS Email (~20,000/month)	£3	Growing user base, more notification triggers
Q4 Monthly Total	£492	

Annual Production Total

Quarter	Monthly	× Months	Subtotal
Q1	£156	3	£468

Quarter	Monthly	× Months	Subtotal
Q2	£256	3	£768
Q3	£371	3	£1,113
Q4	£492	3	£1,476
Annual Production			£3,825

Note on Azure Functions Y1 Consumption: All three production functions (media processing, AWIN reconciliation, payout confirmation) are queue-triggered or timer-triggered and run asynchronously. Cold starts of 10–30 seconds add no user-visible latency. Timer-triggered jobs (earnings confirmation, AWIN reconciliation) are unaffected by cold starts. If sustained high-frequency media processing causes cold starts to degrade the creator upload experience noticeably, upgrading to EP1 Elastic Premium (~£122/month for 1 always-ready instance) can be revisited — but this is unlikely to be needed in year 1.

Part 3: Stripe Connect — Creator Payout Fees

Stripe fees are not infrastructure costs but are operational costs directly linked to creator payout activity. They are included here for completeness.

Stripe Connect Express (UK pricing):

- **£2/active connected account/month** (an account is "active" if it received a payout in that month)
- **0.25% per transfer** to a connected account (minimum £1.50 per transfer)

Quarter	Active Creators (est.)	Account Fees	Transfer Fees	Monthly Total	Quarterly Total
Q1	0 (payouts not live)	£0	£0	£0	£0
Q2	0 (payouts not live)	£0	£0	£0	£0
Q3	~30	£60	£45 (min £1.50 × 30)	£105	£315
Q4	~75	£150	£113 (min £1.50 × 75)	£263	£789
Annual Stripe Total					£1,104

These fees come out of LYKE's 30% commission share before any platform profit. At 30 active creators each paying out £80 average, LYKE's gross revenue from that cohort is £720/month; Stripe takes £105, leaving £615.

Part 4: Developer Tooling & Recurring Subscriptions

Service	Cost	Frequency	Annual	Notes
GitHub Team (3 developers)	\$4/user/month	Monthly	£113	Includes 3,000 Actions minutes/month; private repos; required
GitHub Actions — Linux runners	Included	—	£0	Covers Android builds, .NET tests, Docker builds
GitHub Actions — macOS runners (M1)	\$0.08/minute	Per build	See below	Only needed if no Mac device available for iOS
Apple Developer Program	\$99/year	Annual	£78	Required for App Store distribution and TestFlight
Google Play Developer	\$25 one-time	One-time	£20	One-time registration fee
Domain — be-lyke.clothing	~£30/year	Annual	£30	.clothing TLD renewal; SSL is free via Azure managed certificates
Annual Tooling Total			£241	Excluding macOS CI runners

GitHub Actions macOS Runners (iOS builds)

If the team does not own a Mac, iOS builds must run on GitHub's macOS runners:

- M1 macOS runner: \$0.08/minute
- Estimated build time: 20–25 minutes
- Estimated frequency: 20 builds/month (PRs + releases)
- Monthly cost: $20 \times 22 \text{ min} \times \$0.08 = \sim \$35 = \textbf{£28/month} = \textbf{£330/year}$

If a Mac device is owned (see Part 5), this cost is £0.

Part 5: One-Time Setup Costs

Item	Cost	Required?	Notes
Mac Mini M4 (16GB RAM, 256GB SSD)	£540	Yes, if no Mac available	Required for iOS builds (Xcode only runs on macOS). Mac Mini is the lowest-cost option that meets requirements.
MacBook Pro 14" M4 (if portability needed)	£1,799	Alternative to Mac Mini	Higher cost; prefer if developer needs a portable machine
Apple iPhone (for physical device testing)	£0–£700	Recommended	Simulator catches most issues; physical device testing is best practice before App Store submission. Use existing device if available.
Android test device	£0–£300	Recommended	A mid-range device (Pixel 7a ~£300) is sufficient; emulator covers most cases

Item	Cost	Required?	Notes
AWIN publisher registration	£0	Yes	Free for publishers
Azure initial resource provisioning	£0	Yes	Covered by pay-as-you-go; no setup fee
Terraform remote state storage	~£1/month	Yes	Azure Storage blob for state file; negligible
One-time total (Mac Mini path)	£540–£1,540		Varies based on device testing requirements

Year 1 Cost Summary

Azure Infrastructure

Environment	Annual Cost
Dev/Staging	£501
Production	£3,825
Azure Total	£4,326

All Costs Combined

Category	Annual	Notes
Azure — Dev/Staging	£501	Stable throughout year
Azure — Production	£3,825	Grows from £156 to £492/month
Stripe Connect fees	£1,104	Variable; only from Q3 when payouts go live
GitHub Team plan	£113	Fixed
Apple Developer Program	£78	Fixed annual
Google Play (one-time)	£20	One-time only
Domain renewal	£30	Fixed annual
Recurring Annual Total	£5,671	
Mac Mini M4 (one-time)	£540	If no Mac owned
Additional test devices	£0–£1,000	As needed
GitHub Actions macOS runners	£330	Only if no Mac; otherwise £0
One-time / conditional costs	£540–£1,870	

Category	Annual	Notes
Year 1 Grand Total (Mac Mini path)	~£6,211	
Year 1 Grand Total (MacBook path)	~£7,470	
Year 1 if Mac already owned	~£5,671	

Monthly Progression

Month	Dev/Staging	Production	Stripe	Other	Monthly Total
1–3	£32	£156	£0	£20	£208
4–6	£45	£256	£0	£20	£321
7–9	£45	£371	£105	£20	£541
10–12	£45	£492	£263	£20	£820

"Other" = monthly share of annual subscriptions (GitHub, Apple, domain = £241/year ÷ 12 ≈ £20/month).

Scaling Triggers

These are the specific thresholds that should prompt infrastructure changes. Monitoring alerts should be set on each.

Trigger	Metric	Action	Estimated cost impact
PostgreSQL CPU > 70% sustained	Azure Monitor: cpu_percent	Upgrade D2s_v3 → D4s_v3	+£77/month
PostgreSQL storage > 80%	Azure Portal: storage %	Increase to 128GB or 256GB	+£8–16/month
PostgreSQL read latency > 20ms p95	App Insights custom metric	Add read replica	+£72/month (D2s_v3 replica)
Redis memory > 70% (C1 = 1GB)	Azure Monitor: used_memory	Upgrade C1 → C2 (6GB)	+£70/month
Container App avg CPU > 60%	Azure Monitor: CpuPercentage	Add 1 replica (auto-scales) or upgrade vCPU	+£57–114/month
Container App p95 response time > 500ms	App Insights: requests/duration	Investigate; likely DB or N+1 query before scaling	Varies
Log Analytics ingestion > 8GB/month	Azure Portal billing	Review log verbosity; move verbose logs to cheaper tier	+£8/month per GB

Trigger	Metric	Action	Estimated cost impact
Functions cold start impacting creator uploads	App Insights: media processing duration p95	Upgrade Y1 → EP1 (Elastic Premium, 1 always-ready instance)	+£122/month

Cost Optimisation Opportunities

The following changes are not implemented yet but would reduce costs materially:

1. ☒ Functions: Downgraded EP1 → Y1 Consumption (saving ~£1,488/year) — IMPLEMENTED

Production Functions now use the Y1 Consumption plan instead of EP1 Elastic Premium. All three functions (media processing, AWIN reconciliation, payout confirmation) are queue-triggered or timer-triggered and run asynchronously. Cold starts of 10–30 seconds add no user-visible latency. This saves approximately £119–138/month depending on the quarter. Monitor cold start impact on creator upload experience; upgrade to EP1 only if needed.

2. Redis: Defer to Q2 (~£63/month saving in Q1)

Redis is planned but not yet provisioned. The app currently functions without it (in-memory caching for lookups). Defer provisioning until feed performance degrades under real traffic, rather than provisioning speculatively from day one.

3. CDN: Defer to Q2 (~£8/month saving in Q1)

With a small initial user base, the CDN adds complexity and cost without meaningful benefit. Enable once media assets are being loaded by a meaningful number of concurrent users.

4. Container Registry: Downgrade is not possible (Basic is already minimum)

Basic SKU at ~£4/month is the floor. No action needed.

5. Dev environment: Keep scale-to-zero aggressively

The dev Container App is already configured `min_replicas = 0`. Ensure the CI/CD pipeline does not leave dev running continuously — confirm the app spins down after pipeline smoke tests complete.

6. PostgreSQL dev: Consider pausing overnight

Azure PostgreSQL Flexible Server can be stopped (not deleted) manually to save compute costs when not in use. This is a manual operation, but for a pre-launch team not running 24/7 CI, stopping the dev DB overnight saves ~70% of dev DB compute cost (~£10/month). Requires a startup script or scheduled stop/start.

Azure Budgets and Alerts (Recommended)

Set the following Azure Budget alerts before provisioning prod:

Budget	Alert threshold	Alert at
Production — monthly	£500 (Q1–Q2), £800 (Q3–Q4)	80%, 100%, 110%
Dev/Staging — monthly	£80	80%, 100%
Entire subscription — monthly	£1,000	90%, 100%

The 110% alert on prod is critical: it catches unexpected scaling events or misconfigured retry loops (e.g., a Functions job stuck in a retry storm consuming Consumption plan compute).